Status	Finished
Started	Wednesday, 28 October 2026, 8:31 PM
Completed	Wednesday, 28 October 2026, 8:49 PM
Duration	18 mins 12 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Merge Sort
c. Bubble Sort
d. Insertion Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Merging two datasets
b. Arranging data in a specific order
c. Removing duplicates from a list
d. Finding a specific element in a list

The correct answer is: Arranging data in a specific order

Question **3**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Greedy method
b. Divide and conquer
c. Backtracking
d. Dynamic programming

The correct answer is:
Divide and conquer

Question **4**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It slows down the process
b. It makes the data unsorted
c. It complicates the selection of items
d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Merge Sort
b. Linear Search
c. Binary Search
d. Bubble Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. arrange()
b. sort()
c. order()
d. sorted()

The correct answer is: sort()

Question **7**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It has no effect on the analysis process
b. It allows for quicker identification of duplicates
c. It makes the analysis slower
d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Linear Search
b. Binary Search
c. Merge Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **9**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It simplifies the sorting process by using only one pass
b. It only works on small datasets
c. It avoids the need for recursion
d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **10**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Merge Sort
b. Linear Search
c. Quick Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It allows for faster searching
b. It increases the number of comparisons needed
c. It makes the search operation slower
d. It has no effect on the search operation

The correct answer is: It allows for faster searching

Question **12**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. arrange()
b. sorted()
c. order()
d. sort()

The correct answer is: sorted()

Question **13**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data in a single pass
b. Dividing the input into parts, solving each part, and combining the solutions
c. Rearranging data without sorting
d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **14**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Insertion sort
b. Quick sort
c. Heap sort
d. Bubble sort

The correct answer is:
Insertion sort

Question **15**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Decreases the efficiency of algorithms
b. Makes data harder to manage
c. Provides a basis for other algorithms to work efficiently
d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently